3.3 F下，${}^{F}P=[3,2,2]^T$ F的齐次变换阵：${}^{0}T_F=\begin{bmatrix}0&-1&0&10\\1&0&0&20\\0&0&1&1\\0&0&0&1\end{bmatrix}$

(1) 求同一点在基坐标下表示。

$R=\begin{bmatrix}0&-1&0\\1&0&0\\0&0&1\end{bmatrix}$，$P=\begin{bmatrix}10\\20\\1\end{bmatrix}$

由坐标变换 ${}^{0}P=R\,{}^{F}P+P$

代入：$R\,[3,2,2]^T=[-2,3,2]\Rightarrow {}^{0}P=\begin{bmatrix}-2\\3\\2\end{bmatrix}+\begin{bmatrix}10\\20\\1\end{bmatrix}=\begin{bmatrix}8\\23\\3\end{bmatrix}$

(2) 求新坐标系F'的变换阵

绕坐标系y转90°：$R_y(90°)=\begin{bmatrix}0&0&1&0\\0&1&0&0\\-1&0&0&0\\0&0&0&1\end{bmatrix}$

平移x轴20：$T_x(20)=\begin{bmatrix}0&0&1&21\\1&0&0&20\\0&1&0&-10\\0&0&0&1\end{bmatrix}$

$\Rightarrow {}^{0}T_{F'}=\begin{bmatrix}R'&P'\\0&1\end{bmatrix}$

${}^{F'}P=\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}\begin{bmatrix}-13\\3\\13\end{bmatrix}=\begin{bmatrix}3\\13\\-13\end{bmatrix}$

$\Rightarrow$ 得到：${}^{F'}P=[3,13,-13]^T$

3.4 已知齐次变换：$H=\begin{bmatrix}0&1&0&0\\0&0&-1&0\\-1&0&0&0\\0&0&0&1\end{bmatrix}$

要求确定 $Rot(f,\theta)=H$ 旋转轴f与转角θ

平移为0 $\Rightarrow R=\begin{bmatrix}0&1&0\\0&0&-1\\-1&0&0\end{bmatrix}$

转角θ: trajectory : $1+2\cos\theta$.

$\Rightarrow \leq 0$, $\theta = 120°$

转轴: $f = \frac{1}{2\sin\theta}\begin{bmatrix} r_{32}-r_{23} \\ r_{13}-r_{31} \\ r_{21}-r_{12} \end{bmatrix}$ 代入得: $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

$\because \sin 120° = \sqrt{3}/2 \Rightarrow f = \frac{1}{\sqrt{3}}[1,1,-1]^T$

转轴: $f = \frac{1}{\sqrt{3}}[1,1,-1]^T$

故: $Rot(f,\theta) = Rot\left(\frac{1}{\sqrt{3}}[1,1,-1]^T, 120°\right)$

3.6 PUMA250正运动学

省略DH参数表.

设 $C_i = \cos\theta_i$; $S_i = \sin\theta_i$

相邻DH: $^{i-1}T_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$

连杆变换: $A_1 = {}^0T_1 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$A_2 = {}^1T_2 = \begin{bmatrix} C_2 & -S_2 & 0 & aC_2 \\ S_2 & C_2 & 0 & aS_2 \\ 0 & 0 & 1 & b+d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$A_3 = {}^2T_3 = \begin{bmatrix} C_3 & 0 & S_3 & 0 \\ S_3 & 0 & -C_3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$A_4 = {}^3T_4 = \begin{bmatrix} C_4 & 0 & -S_4 & 0 \\ S_4 & 0 & C_4 & 0 \\ 0 & -1 & 0 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_5 = {}^4T_5 = \begin{bmatrix} C_5 & 0 & -S_5 & 0 \\ S_5 & 0 & C_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_6 = {}^5T_6 = \begin{bmatrix} C_6 & -S_6 & 0 & 0 \\ S_6 & C_6 & 0 & 0 \\ 0 & 0 & 1 & b_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

末端变换阵：${}^0T_6 = A_1 A_2 A_3 A_4 A_5 A_6$

$$ {}^0T_3 = \begin{bmatrix} -C_1 C_{23} & S_1 & C_1 S_{23} & 8C_1C_2 + S_1(b_1 + b_2) \\ S_1 C_{23} & -C_1 & S_1 S_{23} & 8S_1C_2 - C_1(b_1 + b_2) \\ S_{23} & 0 & -C_{23} & 8S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

其中，$C_{23} = \cos(\theta_2 + \theta_3)$，$S_{23} = \sin(\theta_2 + \theta_3)$

$$ {}^3T_6 = \begin{bmatrix} C_4C_5C_6 + S_4S_6 & -C_4C_5S_6 + C_6S_4 & -C_4S_5 & -b_3C_4S_5 \\ -C_4S_6 + C_5C_6C_4 & -C_4C_6 - C_5S_4S_6 & -S_4S_5 & -b_3S_4S_5 \\ -C_6S_5 & S_5S_6 & -C_5 & 8 - b_3C_5 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$${}^0T_6 = {}^0T_3 \, {}^3T_6 = A_1 A_2 A_3 A_4 A_5 A_6$$

3.) $\{B\}$ 基坐标系，$\{W\}$ 世界坐标系，$\{T\}$ 工具坐标系，

$\{S\}$：工件坐标系，$\{G\}$ 目标坐标系，

插入成功时，工具坐标系与目标坐标系重合

故有：${}^BT_T = {}^BT_G \Rightarrow {}^BT_T = {}^B_TW \, {}^W_TT$

故有：$^{B}T_{G} = {}^{B}T_{S}\,{}^{S}T_{G} \Rightarrow {}^{B}T_{W}\,{}^{W}T_{T} = {}^{B}T_{S}\,{}^{S}T_{G}$

$$\Rightarrow {}^{W}T_{T} = \left({}^{B}T_{W}\right)^{-1}\,{}^{B}T_{S}\,{}^{S}T_{G}$$